

ANALYSIS AND SYNTHESIS OF FUNDAMENTAL FREQUENCY CONTOURS FOR THE SPOKEN DIALOGUE IN JAPANESE

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ABSTRACT *Prosodic features of spoken dialogue were analyzed and the results were compared with those of read speech. A wider dynamic range was observed in the fundamental frequencies, accompanied by an increase in their mean value. The wider dynamic range is caused by increases in the amplitude of accent commands and in the magnitude of phrase commands. A wider dynamic range was also observed in the syllable duration. On the average, syllable duration was reduced to a great extent. Based on the results, prosodic rules were constructed for spoken dialogue by modifying those already developed for the read speech. The issue of the focal control was also addressed and incorporated in the prosodic rules. The validity of these rules was proved by the listening test of synthetic speech.*

1. INTRODUCTION

Communication through speech will certainly play a major role in realizing smooth interface between man and machine. Therefore, a large amount of effort is being devoted to construct spoken dialogue systems. Although the realization of such a system requires a wide range of techniques of spoken language processing and dialogue processing, the present paper will deal only with speech synthesis, especially with prosodic control. We have already developed a text-to-speech conversion system for Japanese, and have achieved a very high quality in the prosodic features of synthetic speech using a quantitative model for generating fundamental frequency contours[1]. Since the prosodic features of a spoken dialogue are different from those of text reading, prosodic rules for text-to-speech conversion have to be modified to produce high quality speech for a spoken dialogue system. The present paper describes our analysis of prosodic features of a spoken dialogue, with special emphasis on the fundamental frequency contours (F_0 contours), and compares the results with those obtained for text reading. Based on the comparison, preliminary rules are constructed for the control of prosodic features of synthetic speech to be used in a spoken dialogue system.

2. THE SPEECH MATERIAL AND THE METHOD OF ANALYSIS

Thirteen male and five female speakers of the common Japanese, with some training as actors or actresses, simulated a spontaneous dialogue by referring to a written text of a dialogue on ski resorts. They also uttered individual sentences of the text in a normal reading style. These utterances were recorded and digitized to serve as the material for the analysis of prosodic features. The speech samples in the simulated dialogue shall henceforth be referred to as the dialogue-style samples and those in the normal text reading as the reading-style samples. The results obtained from the analysis of reading-style samples were used as references for dialogue-style samples.

Fundamental frequencies were extracted from the speech samples by detecting the maximum peak of the autocorrelation function of the LPC prediction residual. Unlike conventional pitch extraction methods, this method calculates the autocorrelation function with a frame length proportional to the time lag[2], and gives better results in pitch extraction. The feature parameters of an F_0 contours were then extracted by the method of Analysis-by-Synthesis using a model of F_0 contour generation[3]. This model is based on the assumption that, in the logarithmic scale of frequency, an F_0 contour can be decomposed into phrase and accent components, each being represented by the response of a second-order

critically-damped linear system to the corresponding command. An impulse-like command is assumed for the phrase component, while a step-wise command is assumed for the accent component. The magnitude (A_p) and the timing (T_0) of each phrase command, as well as the amplitude (A_a), the onset timing (T_1) and the offset timing (T_2) of each accent command are major parameters characterizing an F_0 contour.

An analysis was then conducted on the speech rate. Each utterance was segmented into morphemes by visually inspecting the waveform. Although the source power information may play some role in the prosodic control, it was not analyzed in the current study.

3. RESULTS OF ANALYSIS

Table 1 shows the mean (μ) and the standard deviation (σ) of the magnitude of the utterance-initial phrase command A_{p1} , the magnitude of the phrase command other than the utterance-initial A_{pi} , the amplitude of the accent command A_a , the baseline value F_b and the fundamental frequency F_0 . The table also lists the mean and the standard deviation of the speech rate SR . The results are shown individually for the two male speakers YY and TI. Comparison of results for the dialogue-style samples with those for the reading-style samples shows that both μ and σ are larger for F_0 and SR in the dialogue-style samples. The larger value of μ for F_0 is mainly due to an increase in F_b , while the larger value of σ for F_0 is due to an increase in A_p or A_a . The table also shows that the mean values of the magnitude/amplitude parameters are sometimes smaller for the dialogue-style samples. This decrease is accompanied by an increase in F_b .

Figure 1 compares the F_0 contours of a reading-style sample and of a dialogue-style sample with the same phonemic content /naebamade kurumade saNzikaN kakarimasu/ (It will take three hours by car to Naeba.). In the dialogue-style sample, this utterance was preceded by the following utterance pair: /kurumade ikuNdesuka/ (Are you planning to go by car?) and /hai/ (Yes.). The accent command amplitude A_a of the dialogue-style sample is larger for /saNzikaN/, but is smaller for /kurumade/. This example typically illustrates the influence of focus on the dialogue-style utterance as compared with the reading-style utterance, viz., a prominence on /saNzikaN/ and no prominence on /kurumade/. Since the effect of focus is not explicit because of the averaging process in Table 1, the distributions of A_a are shown separately for words to be focused, words to be de-focused, the sentence-final interrogative particle /ka/, and others in Fig. 2.

Proper assignment of focus requires detailed investigations into the relationships between the degree of prominence of a prosodic word (*i.e.*, the relative amplitude of the accent command) and the syntactic, semantic and pragmatic information to be conveyed by an utterance in the framework of a dialogue. From the analysis of the current speech material, however, we assume that a focus is placed on the noun conveying key information in the case of an answer, or on the interrogative pronoun in the case of a question. We also assume that the words already mentioned in a preceding utterance are de-focused by contrast. Thus the above observation shows that the amplitude of the accent command in a dialogue-style utterance is generally higher if a word is focused, and is lower if it is de-focused, as compared with its amplitude in a reading-style utterance.

Table 1. Comparison of the mean (μ) and the standard deviation (σ) of F_0 contour parameter values and speech rates (SR) for the reading-style samples and for the dialogue-style samples.

Speaker		reading-style samples						dialog-style samples					
		A_{p1}	A_{pi} $i > 1$	A_a	F_b [Hz]	F_0 [Hz]	SR [mora/s]	A_{p1}	A_{pi} $i > 1$	A_a	F_b [Hz]	F_0 [Hz]	SR [mora/s]
YY	μ	0.574	0.314	0.428	73.0	148.5	7.3	0.618	0.277	0.548	84.0	202.9	9.7
	σ	0.149	0.118	0.144	9.6	31.2	3.2	0.061	0.135	0.179	7.1	53.4	5.1
TI	μ	0.636	0.270	0.581	58.3	135.1	9.6	0.622	0.342	0.536	70.6	170.7	12.2
	σ	0.095	0.134	0.147	5.9	36.9	2.2	0.204	0.150	0.158	14.5	46.7	5.5

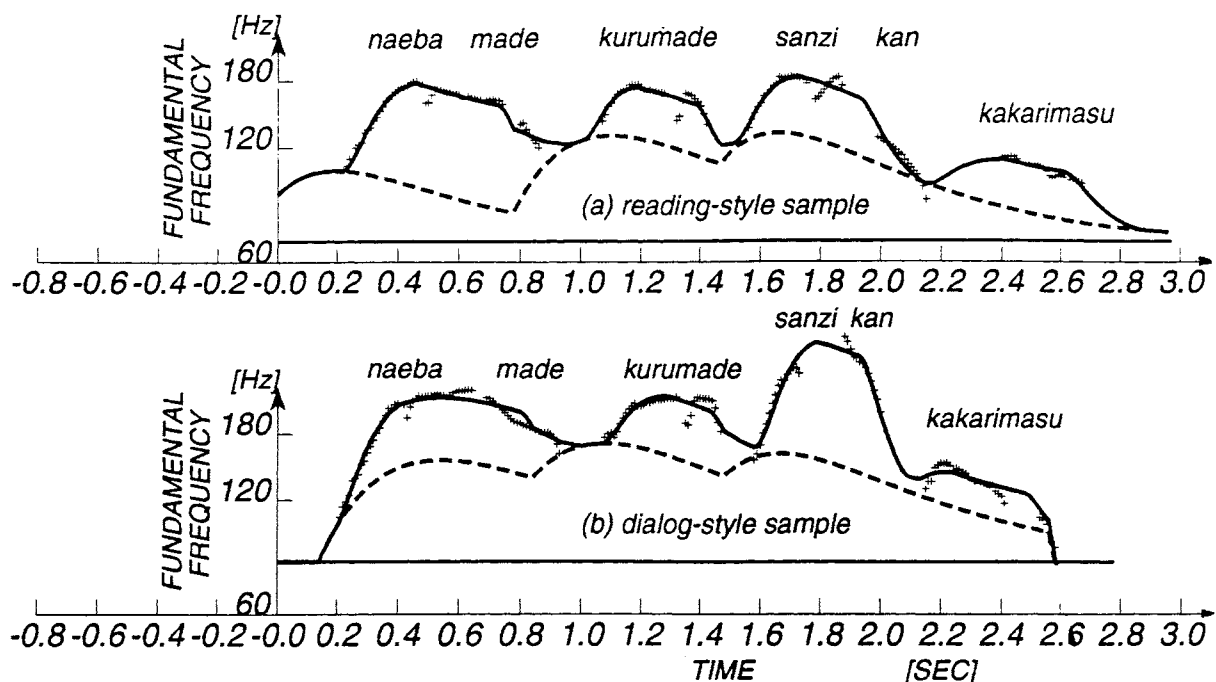


Fig. 1. Results of F_0 contour analysis for the utterance /naebamade kurumade saNzikaN kakarimasu/ (It will take three hours by car to Naeba.): (a) reading-style sample and (b) dialog-style sample.

Although the F_0 rise for the interrogative particle /ka/ can be observed also in reading-style samples, the command timing is somewhat different from that in the dialog-style samples of certain speakers. The onset of the accent command T_1 for dialog-style samples delays by a few ten milli-seconds as compared with the onset for the reading-style samples.

It was also found that, in the case of declarative sentences, the accent command for the predicate phrase usually takes a reduced amplitude toward the end of an utterance.

The larger μ value of SR for the dialog-style samples is accompanied by a larger dynamic range. Figure 3 compares speech rates in morphemes (or their compounds) of both style samples for /hoteruwa tsuiNdake desu ka/ (Does the hotel have only twin-rooms?). The speech rate is kept almost constant in a reading-style sample, while it increases at the middle and decreases toward the end of an utterance in a dialog-style sample. The rate drops markedly at the sentence-final interrogative particle /ka/. A similar tendency is found also in other sentences and in other speakers.

4. SPEECH SYNTHESIS

Based on the results of analysis, prosodic rules were derived for the control of F_0 contours

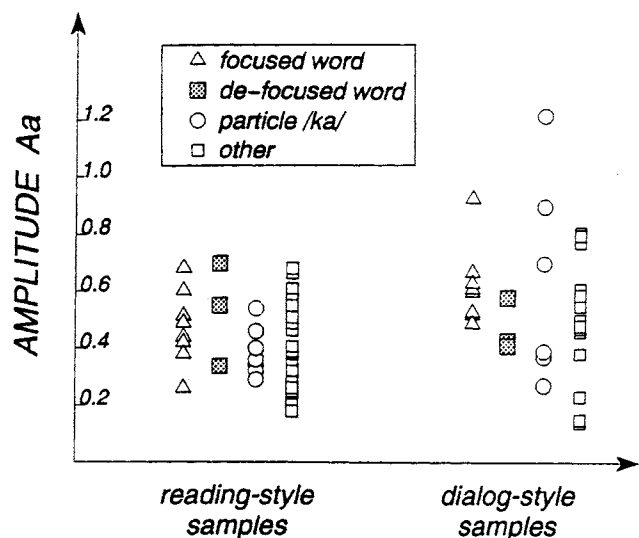


Fig. 2. Comparison of accent command amplitudes for the reading-style samples and for the dialog-style samples uttered by speaker YY. The result is shown separately for focused words, de-focused words, the particle /ka/, and others.

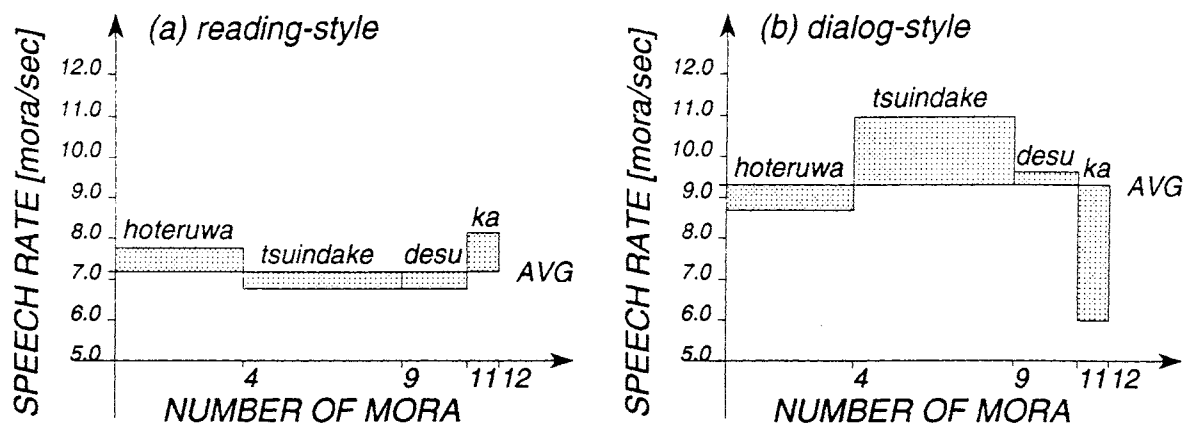


Fig. 3. Speech rates of morphemes (or their compounds) in /hoteruwa tsuindake desu ka/ (Does the hotel have only twin-rooms?): (a) reading-style sample and (b) dialogue-style sample.

of synthetic speech used in the spoken dialogue, by modifying those previously developed for text-to-speech conversion[1]. As for the prosodic symbols, the new rules use the same number of symbols but the values of phrase and accent commands are increased[4]. For instance, the value is set at 0.6 for the accent command FH for the focused prosodic word without apparent downfall in F_0 . The command for the interrogative particle /ka/ is also assigned a large value (*i.e.*, DH=0.6), and its onset timing is delayed by 70 msec as compared with the onset timing of the command for the lexical accent. The rules for generating the prosodic symbols from the linguistic information, developed for the text-to-speech conversion system, are used without modification for the spoken dialogue system. These rules include those dealing with the assignment of prominence.

As for the timing control, the speech rate for the initial prosodic word and the sentence-final auxiliary verb (/desu/ or /masu/) is increased while that of the remaining part is decreased, with an average speech rate set equal to 10 mora/sec.

An experiment on speech synthesis for the dialogue system was conducted using the formant synthesizer with four parallel paths, each consisting of a cascade connection of pole and zero filters[5]. For the sake of comparison, speech samples were also synthesized using the prosodic rules for text-to-speech synthesis. A paired comparison test was carried out on the overall appropriateness as synthetic speech for the spoken dialogue. Speech samples synthesized using the newly developed rules was preferred at a rate of 77.4 %, indicating the validity of the rules.

5. CONCLUSION

In order to construct prosodic rules for speech synthesis to be used in spoken dialogue systems, analysis was conducted on the prosodic features, especially on the F_0 contours of dialogue speech. Based on the results, preliminary rules were derived. For the refinement of the rules, further studies are necessary to find out which features are essential and which features are not essential for the spoken dialogue.

REFERENCES

- [1] K. Hirose and H. Fujisaki, "A system for the synthesis of high-quality speech from texts on general weather conditions," *IEICE Trans. Fundamentals of ECCS*, Vol.E76-A, No.11, pp.1971-1980 (1993-11).
- [2] K. Hirose, H. Fujisaki and S. Seto, "A scheme for pitch extraction of speech using autocorrelation function with frame length proportional to the time lag," *Proc. IEEE ICASSP92*, Vol.I, pp.149-152 (1992-3).
- [3] H. Fujisaki and K. Hirose, "Analysis of voice fundamental frequency contours for declarative sentences of Japanese," *Journal of Acoust. Soc. Jpn.*, Vol.5, No.4, pp.233-242 (1984-10).
- [4] K. Hirose and Y. Asano, "Generation of speech reply in the speech response system," *Proc. EURO-SPEECH93*, Berlin, Vol.2, pp.1319-1322 (1993-9).
- [5] H. Fujisaki, K. Hirose and Y. Asano, "Proposal and evaluation of a new type of terminal analog speech synthesizer," *Proc. ICSLP90*, Kobe, Vol.1, pp.473-476 (1990-10).